

Formula Sheet — Cocoa Volatility Forecasting

Meeting with Dr. Johannes Bleher, 17 March 2026 — updated to current thesis state

1 Volatility Measure: Yang-Zhang (2000)

The Yang-Zhang estimator combines three variance components into a minimum-variance unbiased estimator of daily variance:

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_O^2 + k \hat{\sigma}_C^2 + (1 - k) \hat{\sigma}_{RS}^2$$

The three components are:

Overnight variance:
$$\hat{\sigma}_O^2 = \frac{1}{n-1} \sum_{i=1}^n \left(\ln \frac{O_i}{C_{i-1}} - \ln \frac{\overline{O}}{\overline{C_{-1}}} \right)^2$$

Open-to-close variance:
$$\hat{\sigma}_C^2 = \frac{1}{n-1} \sum_{i=1}^n \left(\ln \frac{C_i}{O_i} - \ln \frac{\overline{C}}{\overline{O}} \right)^2$$

Rogers-Satchell:
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{i=1}^n \left[\ln \frac{H_i}{O_i} \ln \frac{H_i}{C_i} + \ln \frac{L_i}{O_i} \ln \frac{L_i}{C_i} \right]$$

The weighting factor that minimises estimator variance:

$$k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

where n is the estimation window length. The Rogers-Satchell component exploits the daily range (High, Low) for efficiency gains, while the overnight component handles opening gaps. Yang-Zhang achieves 7–8× the efficiency of close-to-close estimators.

Annualisation: $\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252}$ (252 trading days per year; assumes uncorrelated daily returns.)

2 Log Returns

$$r_t = \ln \left(\frac{C_t}{C_{t-1}} \right)$$

where C_t is the closing price on day t . Log returns are time-additive ($r_{t:t+h} = \sum_{i=1}^h r_{t+i}$) and approximately normal over short horizons.

3 Realized Volatility (5-minute, validation only)

$$RV_t = \sum_{i=1}^M r_{t,i}^2$$

where $r_{t,i}$ is the log return for the i -th five-minute interval on day t , and $M \approx 78$ intervals per 6.5-hour trading session. Used only for proxy validation (we have ~ 1 year of intraday data).

4 Proxy Validation: Mincer-Zarnowitz Regression

To validate Yang-Zhang as a proxy for true (unobservable) volatility, we regress realized volatility on Yang-Zhang:

$$RV_t = \alpha + \beta \hat{\sigma}_{YZ,t} + \varepsilon_t$$

Null hypothesis (unbiased proxy): $H_0 : \alpha = 0, \beta = 1$.

Interpretation:

- R^2 measures how much variation in RV is explained by YZ.
 - Our result: $R^2 = 0.46$ at the daily level. This is reasonable — even different intraday sampling frequencies yield $R^2 \approx 0.6$ – 0.7 among themselves.
 - Rolling 22-day averages improve correlation substantially (smooths daily noise). Comparison at the multi-day level aligns with the HAR model's use of averaged volatility as both predictors and targets.
 - Joint F -test for $(\alpha = 0, \beta = 1)$ tests formal unbiasedness.
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5 HAR Components (Corsi, 2009)

The Heterogeneous Autoregressive model decomposes information flow at three frequencies — daily, weekly, monthly — reflecting the behaviour of different market participants:

$$\begin{aligned}\sigma_t^{(d)} &= \sigma_t && \text{(daily — 1 day)} \\ \sigma_t^{(w)} &= \frac{1}{5} \sum_{i=0}^4 \sigma_{t-i} && \text{(weekly — 5-day average)} \\ \sigma_t^{(m)} &= \frac{1}{22} \sum_{i=0}^{21} \sigma_{t-i} && \text{(monthly — 22-day average)}\end{aligned}$$

Forecast target (average volatility over horizon h , used as the dependent variable in direct projection):

$$\bar{\sigma}_{t:t+h} = \frac{1}{h} \sum_{i=1}^h \sigma_{t+i}$$

This averaging ensures comparability across horizons. At $h = 1$ it reduces to tomorrow's volatility; at $h = 264$ it is the average daily volatility over the next 12 months.

6 HAR-QR Model — Main Model (RQ1)

The HAR-QR model estimates *conditional quantiles* of future average volatility, with separate coefficients for each quantile τ :

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau) \sigma_t^{(d)} + \beta_w(\tau) \sigma_t^{(w)} + \beta_m(\tau) \sigma_t^{(m)}$$

where $\mathcal{F}_t$ is the information set at time t .

Key properties:

- Each quantile τ has *its own* coefficient vector $\beta(\tau) = (\beta_0(\tau), \beta_d(\tau), \beta_w(\tau), \beta_m(\tau))'$.
- At $\tau = 0.50$: the conditional median (comparable to OLS mean).
- At $\tau = 0.95$: the conditional upper bound — primary interest for upper-tail risk.
- Quantiles estimated: $\tau \in \{0.50, 0.75, 0.95\}$.
- Horizons: $h \in \{1, 5, 22, 66, 132, 264\}$ trading days (1d, 1w, 1m, 3m, 6m, 12m).
- Separate model for each (h, τ) pair $\Rightarrow 6 \times 3 = 18$ models.

7 HAR-X-QR Model with ENSO — Extended Model (RQ2)

The HAR-X-QR augments HAR-QR with the ENSO climate index:

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau) \sigma_t^{(d)} + \beta_w(\tau) \sigma_t^{(w)} + \beta_m(\tau) \sigma_t^{(m)} + \gamma(\tau) \text{ENSO}_t$$

ENSO variable:

- ENSO_t = Niño 3.4 sea surface temperature anomaly (monthly, NOAA ERSSTv5; anomaly relative to 1991–2020 climatological mean)
- Positive anomalies $\Rightarrow$ El Niño; negative anomalies $\Rightarrow$ La Niña
- Enters *contemporaneously*: ENSO_t predicts $\bar{\sigma}_{t:t+h}$. Fully causal — monthly Niño 3.4 is publicly observable before any forecast is issued.
- Each trading day inherits the Niño 3.4 anomaly of its calendar month.

Transmission mechanism (hypothesis): El Niño $\rightarrow$ drought in West Africa (Sep–Dec) $\rightarrow$ reduced cocoa yields $\rightarrow$ supply disruption $\rightarrow$ volatility spike 6–12 months later.

8 Robustness Diagnostics for the ENSO Signal

Three diagnostics test whether the long-horizon HAR-X-QR gain is a stable structural signal or a sample-specific artefact. Each addresses an orthogonal threat to validity.

(a) Temporal stability — pre-crisis subsample. The sup-Wald structural break test (Andrews 1993) applied to out-of-sample Yang-Zhang volatility identifies **8 February 2024** as the single most significant break (sup-Wald $F = 1,809$, well above the 1% critical value of 10.17), corresponding to a $2.3\times$ step-increase in mean daily volatility. HAR-QR and HAR-X-QR quantile losses are re-evaluated restricting to forecast origins t with $t + h < 8 \text{ Feb } 2024$ ($N \approx 10,076$ pairs). If the gain vanishes in the pre-crisis period, it was regime-driven rather than structural.

Result: At 12m, the pre-crisis improvement retains 79% of the full-sample gain (21.4% vs. 25.4%); at 6m, 57% is retained (9.5% vs. 14.7%). The 3m gain vanishes entirely (−0.4%). The long-horizon signal partially pre-dates the crisis, but the crisis amplifies it.

(b) Lag-decay — physical-channel test. Replace ENSO_t with ENSO_{t-k} for $k \in \{0, 1, 3, 6\}$ months. Under the agronomic-transmission hypothesis, predictive power should *strengthen* at lags 6–12 months (the window over which SST anomalies affect harvests). Monotone decay as k grows contradicts this hypothesis.

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d \sigma_t^{(d)} + \beta_w \sigma_t^{(w)} + \beta_m \sigma_t^{(m)} + \gamma(\tau) \text{ENSO}_{t-k}$$

Result: At 12m, predictive power peaks at $k = 1$ and decays sharply at $k = 3$ and $k = 6$ (which nearly matches the no-ENSO baseline). This is inconsistent with the physical channel and is instead consistent with ENSO acting as a contemporaneous regime-state indicator. Confirmed by cross-correlogram: significant correlations only at lags 0–2, not at the predicted 6–12 month window.

(c) Cross-commodity replication. Apply the identical HAR-X-QR specification to Arabica coffee (ICE KC=F) and raw sugar (ICE SB=F) — tropical soft commodities sharing weather exposure and downstream markets with cocoa. A structural climate channel should produce comparable gains across all three.

Result: ENSO augmentation *worsens* coffee forecasts at both horizons and produces a gain 22× smaller for sugar than for cocoa at 12m. The gain is specific to cocoa’s sample period, not a general tropical-commodity climate signal.

(d) ENSO volatility operationalization. Replace ENSO_t level with a rolling standard deviation of Niño 3.4: sd_{12m} (12-month window) and sd_{36m} (36-month window). Motivated by the hypothesis that switching frequency between ENSO phases drives disruption rather than their level.

Result: At (12m, $\tau = 0.95$), vol-12m achieves 94% of the level specification’s gain; vol-36m only 25%. In the pre-crisis subsample, vol-12m retains 91% of its gain (comparable to level); vol-36m nearly vanishes (18% retention). Vol-36m is fragile; vol-12m mirrors the level’s partial stability.

9 Estimation: Check Function (Pinball Loss)

Quantile regression minimises the asymmetric check function loss:

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_\tau(\bar{\sigma}_{t:t+h} - \mathbf{X}_t' \beta)$$

where $\mathbf{X}_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)})'$ for HAR-QR, or $\mathbf{X}_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)}, \text{ENSO}_t)'$ for HAR-X-QR.

Check function:

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{u < 0}) = \begin{cases} \tau \cdot u & \text{if } u \geq 0 \quad (\text{under-prediction}), \\ (\tau - 1) \cdot u & \text{if } u < 0 \quad (\text{over-prediction}). \end{cases}$$

The minimisation is a linear programming problem (Koenker & d’Orey 1987), solved by the simplex algorithm. Standard errors via bootstrap.

10 Benchmark Models

(a) Historical Volatility (rolling 252-day window, non-parametric):

$$\hat{\sigma}_{t+h}^{\text{hist}} = \sqrt{\frac{1}{252} \sum_{i=1}^{252} (r_{t-i+1} - \bar{r})^2}$$

Quantile forecasts: empirical quantiles of the past 252 volatility observations (sort and select at position $\lceil \tau \cdot 252 \rceil$). Captures only unconditional variation — naive baseline.

(b) GARCH(1,1) (Bollerslev, 1986). Conditional-variance recursion:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Closed-form multi-step forecast:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2), \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}.$$

At long horizons $(\alpha + \beta)^{h-1} \rightarrow 0$ exponentially, so all multi-step forecasts collapse to $\bar{\sigma}^2$ — the *long-horizon degeneration* of daily-GARCH. For $\alpha + \beta \approx 0.97$, at $h = 264$: $(0.97)^{263} \approx 0.0003$.

Horizon- h point forecast: $\hat{\sigma}_{t,h} = \sqrt{h^{-1} \sum_{k=1}^h \mathbb{E}_t[\sigma_{t+k}^2]}$.

Quantile forecasts (Gaussian closed-form, consistent with normal-innovation assumption):

$$\hat{Q}_{\tau,t}^{\text{GARCH}} = \hat{\sigma}_{t,h} + s_t \cdot \Phi^{-1}(\tau),$$

where s_t is the in-estimation-window standard deviation of σ_{YZ} (proxy for forecast uncertainty) and Φ^{-1} is the standard-normal inverse CDF.

Units note: `arch` fit on $r_t \times 100$ for numerical stability, so ω is in (%)². All parameters rescaled to fraction² units ($\omega^* = \omega/10^4$) before the recursion to ensure consistent units throughout.

(c) HAR-OLS (conditional mean + pseudo-quantile):

$$\mathbb{E}[\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t] = \beta_0 + \beta_d \sigma_t^{(d)} + \beta_w \sigma_t^{(w)} + \beta_m \sigma_t^{(m)}$$

Estimated by OLS (minimises $\sum (y - \hat{y})^2$). For distributional forecasts, let $\hat{\varepsilon}_i = \sigma_i - \hat{\mu}_i$ denote the training-window residuals. The τ -quantile forecast is:

$$\hat{Q}_{\tau,t}^{\text{OLS}} = \hat{\mu}_{t,h} + \hat{F}_{\varepsilon}^{-1}(\tau),$$

where $\hat{F}_{\varepsilon}^{-1}(\tau)$ is the empirical τ -quantile of the training-window residuals. Requires no distributional assumption. Tests whether explicit quantile regression adds value beyond conditioning on the HAR structure alone.

11 Evaluation Metrics

(a) QLIKE — for mean/median forecasts (robust to proxy noise, Patton 2011):

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\ln \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right) \quad (\text{more negative} = \text{better})$$

Patton (2011): among standard loss functions, only MSE and QLIKE produce model rankings robust to noise in the volatility proxy. QLIKE penalises under-prediction proportionally more than MSE.

(b) MSE and MAE — point-forecast robustness checks:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\sigma_t - \hat{\sigma}_t)^2, \quad \text{MAE} = \frac{1}{T} \sum_{t=1}^T |\sigma_t - \hat{\sigma}_t|.$$

MSE is proxy-robust (same class as QLIKE); MAE is not proxy-robust but gives an interpretable error in original volatility units. Both confirm QLIKE rankings hold across loss functions.

(c) Quantile Loss — for distributional forecasts:

$$\text{QL}_{\tau} = \frac{1}{T} \sum_{t=1}^T \rho_{\tau}(\sigma_t - \hat{Q}_{\tau,t}) \quad (\text{lower} = \text{better})$$

Uses the same check function ρ_{τ} as estimation — HAR-QR is *coherent*: estimation and evaluation loss functions are aligned.

(d) Empirical Violation Rate — calibration diagnostic:

$$\widehat{VR}_{\tau} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\sigma_t > \hat{Q}_{\tau,t}\}$$

A well-calibrated τ -quantile forecast should have $\widehat{VR}_{\tau} \approx 1 - \tau$.

Results (HAR-QR, $\tau = 0.95$):

Horizon	N	Violations	$\widehat{VR}$ (%)
1d	10,545	510	4.84
1w	10,541	543	5.15
1m	10,524	600	5.70
3m	10,480	618	5.90
6m	10,414	733	7.04
12m	10,282	857	8.34

HAR-X-QR at 12m: $\widehat{VR} = 10.29\%$ — model positions the boundary more aggressively in El Niño periods, incurring more breaches but achieving lower quantile loss overall.

(e) Model Confidence Set (MCS) — Hansen, Lunde & Nason (2011):

Sequential testing for comparing multiple models simultaneously. Iteratively eliminates the worst-performing model until the null of equal predictive ability is not rejected. Surviving set $\mathcal{M}^*$ contains the best model at $(1 - \alpha)$ confidence.

Our settings: $\alpha = 0.10$ (90% confidence), 10 000 stationary bootstrap replications, max- t statistic. Five models evaluated: Historical Vol, GARCH(1,1), HAR-OLS, HAR-QR, HAR-X-QR.

Interpretation of p -values:

- $p = 1.00$: last model eliminated = best performer
- $p < 0.10$: excluded from $\mathcal{M}^*$ (significantly worse)
- $p \in [0.10, 1.00)$: included in $\mathcal{M}^*$ but not the single best

Why MCS over Diebold-Mariano? DM is pairwise. With 5 models, that requires 10 pairwise tests — multiple testing problem. MCS handles the joint comparison and controls the familywise error rate.

12 Stationarity Tests

Augmented Dickey-Fuller (ADF):

$$\Delta y_t = \alpha + \delta y_{t-1} + \sum_{j=1}^p \phi_j \Delta y_{t-j} + \varepsilon_t$$

H_0 : $\delta = 0$ (unit root / non-stationary). Lag length p selected by AIC.

Phillips-Perron (PP): Same null, non-parametric Newey-West HAC correction. More robust to serial correlation and heteroskedasticity.

Our results: All series reject H_0 at 1%:

Series	N	ADF stat	PP stat	p -value
Log returns	13,087	−112.08	−112.09	< 0.001
Yang-Zhang vol.	13,087	−8.42	−90.99	< 0.001
Niño 3.4	787	−6.85	−6.46	< 0.001

Note: The ADF/PP discrepancy for YZ volatility (−8.42 vs −90.99) is because ADF needs 39 lags to whiten the highly persistent residuals, while PP uses a non-parametric correction. Both agree on rejection.

13 Key Setup Parameters

Rolling estimation window:	2 520 trading days $\approx$ 10 years
Out-of-sample forecasts:	$\approx$ 10 500 daily forecast origins
Re-estimation frequency:	daily rolling (1-day step)
Data:	ICE London Cocoa (LCCc1), daily OHLC+V, 1984–2025
Horizons:	$h \in \{1, 5, 22, 66, 132, 264\}$ days
Quantiles:	$\tau \in \{0.50, 0.75, 0.95\}$
Direct projection:	Separate model per (h, τ) pair
Total models:	5 model types (Hist. Vol., GARCH, HAR-OLS, HAR-QR, HAR-X-QR) $\times$ 6 horizons $\times$ 3 quantiles
Structural break date:	8 February 2024 (sup-Wald, Andrews 1993)
Pre-crisis subsample:	$N \approx 10,076$ pairs (before 8 Feb 2024)